

gg: (h-10)

(h-3) Frequency Response

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$$\text{Magnitude} = 20 \log [1 + r_1^2 - 2r_1 \cos(\omega - \theta_1)] \leftarrow \text{zero} \rightarrow +$$

$$+ 10 \log [1 + r_1^2 - 2r_1 \cos(\omega + \theta_1)]$$

$$- 10 \log [1 + r_2^2 - 2r_2 \cos(\omega - \theta_2)] \leftarrow \text{pole} \rightarrow -$$

$$- 10 \log [1 + r_2^2 - 2r_2 \cos(\omega + \theta_2)] \leftarrow \text{pole} \rightarrow -$$

Phase / Argument:

$$\angle H(j\omega) = 20 \log |H(j\omega)|$$

$$= \tan^{-1} \left[\frac{r_1 \sin(\omega - \theta_1)}{1 - r_1 \cos(\omega - \theta_1)} \right] + \tan^{-1} \left[\frac{r_1 \sin(\omega + \theta_1)}{1 - r_1 \cos(\omega + \theta_1)} \right] \leftarrow \text{zero}$$

$$- \tan^{-1} \left[\frac{r_2 \sin(\omega - \theta_2)}{1 - r_2 \cos(\omega - \theta_2)} \right] + \tan^{-1} \left[\frac{r_2 \sin(\omega + \theta_2)}{1 - r_2 \cos(\omega + \theta_2)} \right] \leftarrow \text{pole}$$

so,

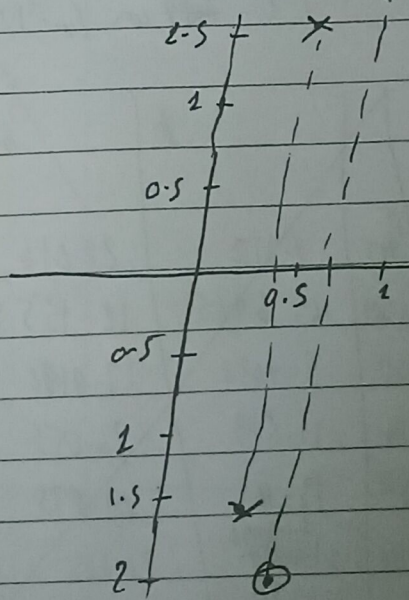
82 Bb

$$p = 0.45 \pm j 2.6 \leftarrow r_2$$

$$z = 0.58 \pm j 2.06 \leftarrow r_1$$

$$r_1 = 2.14 < \pm 1.296$$

$$r_2 = 1.66 < \pm 1.296$$



so,

$$\text{Magnitude} = 20 \log [1 + r_1^2 - 2r_1 \cos(\omega - \theta_1)] + 10 \log [1 + r_2^2 - 2r_2 \cos(\omega + \theta_2)]$$

$$- 10 \log [1 + r_2^2 - 2r_2 \cos(\omega - \theta_2)] - 10 \log [1 + r_1^2 - 2r_1 \cos(\omega + \theta_1)]$$

$$= 10 \log [1 + 2.14^2 - 2 \times 2.14 \cos(\omega - 1.296)] + 10 \log [1 + 1.66^2 - 2 \times 1.66 \cos(\omega + 1.296)]$$

$$- 10 \log [1 + 1.66^2 - 2 \times 1.66 \cos(\omega - 1.296)] - 10 \log [1 + 2.14^2 - 2 \times 2.14 \cos(\omega + 1.296)]$$

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1.296

$$\text{Mag} = 10 \log [5.5796 - 4.28 \cos(\omega - 0.1)] + 10 \log [5.5796 - 4.28 \cos(\omega + 0.1)]$$

$$- 10 \log [3.7556 - 3.32 \cos(\omega - 1.296)] - 10 \log [3.7556 - 3.32 \cos(\omega + 1.296)]$$

↓

b

Argument: $\tan^{-1} \left[\frac{r_1 \sin(\omega - \theta_1)}{1 - r_1 \cos(\omega - \theta_1)} \right] + \tan^{-1} \left[\frac{r_2 \sin(\omega + \theta_1)}{1 - r_2 \cos(\omega + \theta_1)} \right]$

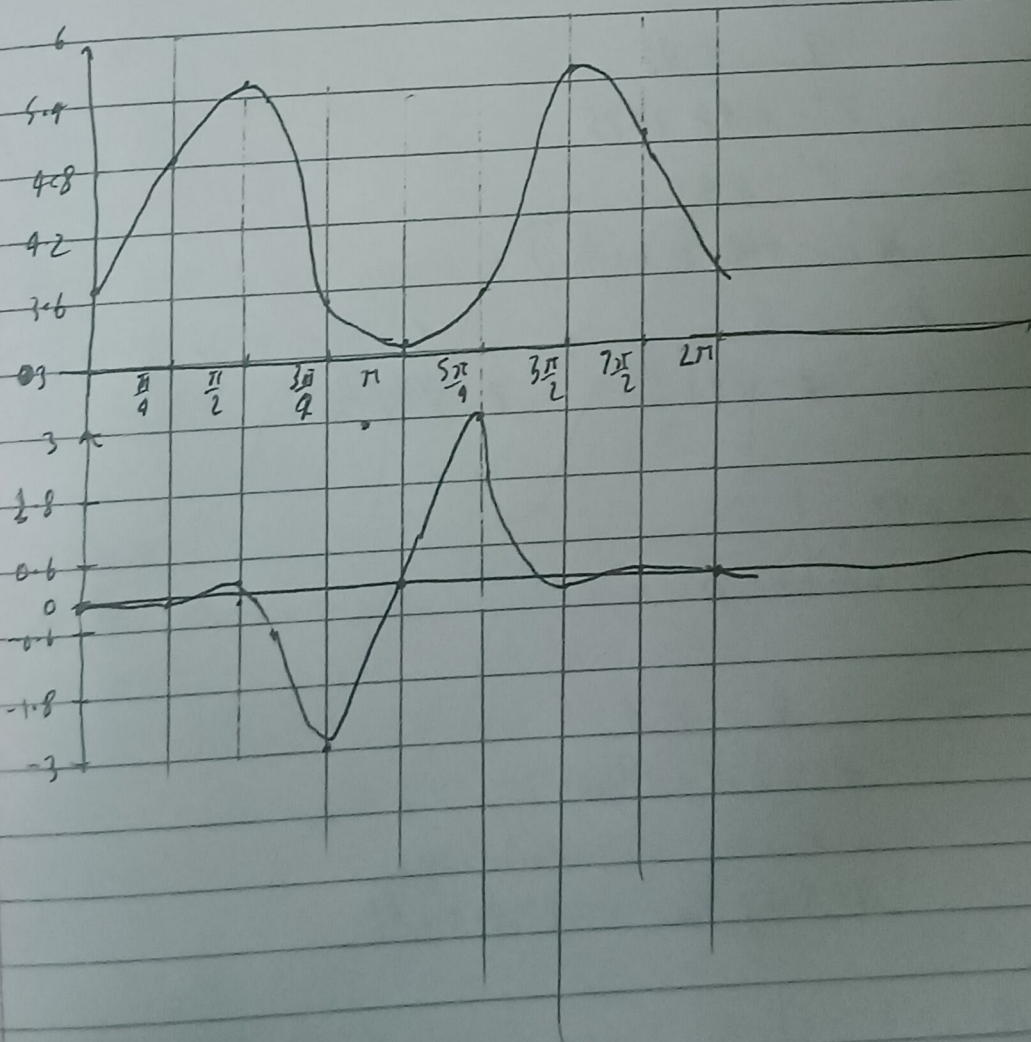
$- \tan^{-1} \left[\frac{r_2 \sin(\omega - \theta_2)}{1 - r_2 \cos(\omega - \theta_2)} \right] - \tan^{-1} \left[\frac{r_2 \sin(\omega + \theta_2)}{1 - r_2 \cos(\omega + \theta_2)} \right]$

$\phi = \tan^{-1} \left[\frac{2.19 \sin(\omega - 1.296)}{1 - 2.19 \cos(\omega - 1.296)} \right] + \tan^{-1} \left[\frac{2.19 \sin(\omega + 1.296)}{1 - 2.19 \cos(\omega + 1.296)} \right]$

$\phi = - \tan^{-1} \left[\frac{2.66 \sin(\omega - 1.296)}{1 - 2.66 \cos(\omega - 1.296)} \right] - \tan^{-1} \left[\frac{2.66 \sin(\omega + 1.296)}{1 - 2.66 \cos(\omega + 1.296)} \right]$

ω	a	b	mag	c	d	ϕ
0	12.604	9.112	22.016	0	0	0
$\pi/4$	12.570	6.6465	18.1555	1.553	1.7409	
$\pi/2$	11.509	5.9036	17.9196	0.329	-0.979	
$3\pi/4$	10.750	11.519	26.655	-1.866	1.1212	
π	11.511	13.361	29.825	0.825	0	
$5\pi/4$	13.385	15.116	26.655	-1.896	-1.212	
$3\pi/2$	15.116	11.511	17.9196	0.3137	0.979	
$7\pi/8$	16.248	6.6465	18.1555	-1.805	-1.79	
2π	16.574	9.1112	22.016	0	0	

ω	a	b	mag	c	d	$\angle$
0	12.904	9.112	3.792	0	0	0
$\pi/4$	11.509	6.1465	4.8625	1.6183	1.7404	-0.1221
$\pi/2$	11.511	5.9036	5.6074	-0.3137	-0.979	0.1603
$3\pi/4$	15.116	11.519	3.597	-1.896	1.121	-0.7817
π	16.514	13.361	3.153	0	0	-3.017
$5\pi/4$	15.216	11.519	3.597	1.846	-1.121	3.017
$3\pi/2$	11.511	5.9036	5.6074	0.3137	0.9791	-0.1603
$7\pi/4$	11.509	6.1465	4.8625	-1.6183	-1.7404	0.1221
2π	12.904	9.112	3.792	0	0	0



if difference eqⁿ given:

8/13n:

$$y[n] - 0.4y[n-1] + 0.25y[n-2] = x[n] - 0.9x[n-1]$$

ZT:

$$Y(z) - 0.4z^{-1}Y(z) + 0.25z^{-2}Y(z) = X(z) - 0.9z^{-1}X(z)$$

$$\begin{aligned} \therefore \frac{Y(z)}{X(z)} &= \frac{1 - 0.9z^{-1}}{1 - 0.4z^{-1} + 0.25z^{-2}} \\ &= \frac{z^2 - 0.9z}{-1z + 0.1} \\ &= \frac{z^2 - 0.9z}{z^2 - 0.4z + 0.25} \end{aligned}$$

$$\begin{aligned} \frac{H(z)}{z} &= \frac{z^2 - 0.9z}{(z)(z - \frac{1}{5} \pm j\frac{\sqrt{11}}{10})} \\ &= \frac{A}{z} + \frac{B}{z - \frac{1}{5} + j\frac{\sqrt{11}}{10}} + \frac{C}{z - \frac{1}{5} - j\frac{\sqrt{11}}{10}} \end{aligned}$$

$$A = \frac{z^2 - 0.9z}{z^2 - 0.4z + 0.25} \Big|_{z=0}$$

$$\begin{aligned} \frac{H(z)}{z} &= \frac{z - 0.9}{z - \frac{1}{5} \pm j\frac{\sqrt{11}}{10}} \\ &= \frac{A}{z - \frac{1}{5} + j\frac{\sqrt{11}}{10}} + \frac{A^*}{z - \frac{1}{5} - j\frac{\sqrt{11}}{10}} \end{aligned}$$

$$\begin{aligned} A &= \frac{z - 0.9}{z - \frac{1}{5} - j\frac{\sqrt{11}}{10}} \Big|_{z = \frac{1}{5} - j\frac{\sqrt{11}}{10}} \\ &= 0.718 + j 0.282 \end{aligned}$$

$$\therefore A^* = 0.718 - j 0.282$$

$$H(z) = \frac{(0.718 + j0.282)z}{z - \frac{1}{2} + j\frac{\sqrt{2}}{10}} + \frac{(0.718 - j0.282)z}{z - \frac{1}{2} - j\frac{\sqrt{2}}{10}}$$

$$\begin{aligned} H(n) &= (0.718 + j0.282) \left(\frac{1}{2} - j\frac{\sqrt{2}}{10}\right)^n + (0.718 - j0.282) \left(\frac{1}{2} + j\frac{\sqrt{2}}{10}\right)^n \\ &= (0.718 + j0.282) \left[\frac{1}{2}\right]^n \left[\cos(n \times 1.59) - j \sin(n \times 1.59) \right] \\ &\quad + (0.718 - j0.282) \left[\frac{1}{2}\right]^n \left[\cos(1.59n) + j \sin(1.59n) \right] \\ &= \left[\frac{1}{2}\right]^n \left[\begin{array}{l} 0.718 \cos(1.59n) + 0.718 \cos(1.59n) - j0.718 \sin(1.59n) \\ + j0.282 \cos(1.59n) - j0.282 \cos(1.59n) + j0.718 \sin(1.59n) \\ + j0.282 \sin(1.59n) + j0.282 \sin(1.59n) \end{array} \right] \\ &= \left[\frac{1}{2}\right]^n \left[1.436 \cos(1.59n) + 0.564 \sin(1.59n) \right] \end{aligned}$$

maile eho kina garako? why? k kam thyo?
sida yogaru

$$H(z) = \frac{z^2 - 0.4z}{z^2 - 0.4z + 0.25} = \frac{z(z - 0.4)}{z^2 - 0.4z + 0.25}$$

zeros:

$$z_1 = 0, z_2 = 0.4$$

for poles:

$$z^2 - 0.4z + 0.25 = 0$$

$$z = 0.2 \pm j0.458 \in p_1, p_2$$

solve normally for poles.

use formula for individual zero.

$$10 \log [1 + r^2 - 2r \cos(\omega - \theta)] \times 2 \text{ for } z_1, z_2 \text{ separately.}$$